

VISHNU P

AI Engineer · Generative AI · Agentic Systems

+91-7736533119 · vishnup22102002@gmail.com · linkedin.com/in/vishnup22102002 · github.com/vishnup102002 · vishnup.vercel.app · Tirur, Kerala

PROFILE SUMMARY

Systems-oriented AI Engineer and MCA Graduate with hands-on experience in Generative AI and Agentic orchestration. Designed and deployed production-grade multi-agent pipelines, real-time collaborative systems, and cross-platform applications. Brings R&D internship experience and a strong project portfolio to a junior AI Engineer role.

EXPERIENCE

R&D Intern — AI / Agentic Systems — Bluecast Technologies

Jan 2026 – Apr 2026

- Built a DXF-to-JSON agentic pipeline using ReAct-style loops and GPT-4o to automate indoor navigation mapping from architectural drawings.
- Designed a rule-cascade system — geometric algorithms extract spatial data first; LLM is called only when rules cannot resolve ambiguity — cutting API costs by 75%.
- Delivered schema-validated JSON outputs enabling real-time navigation with zero client-side post-processing.

PROJECTS

Career-Pilot — Autonomous AI Job-Hunting Agent

Python · LangGraph · FastAPI · MCP · Docker

- Built a 4-agent pipeline (Research → Scout → Writer → Alert) using LangGraph with conditional exits and state checkpointing; scrapes job listings via a FastMCP + Playwright server and matches them to the resume in a single batched Groq API call.
- Deduplicates seen URLs via SQLite, scores each job out of 5, and dispatches a daily digest via SendGrid. Deployed on Hugging Face Spaces via Docker.
- Live: huggingface.co/spaces/Vishnuporkulath/career-pilot

GestureLearn — Gesture-Based Collaborative Platform

React · WebRTC · Socket.io · MediaPipe · Node.js

- Architected multi-peer WebRTC (RTCPeerConnection) for real-time audio/video streaming, paired with Socket.io relay to sync canvas strokes and state across peers with sub-50ms latency.
- Built a deterministic gesture state machine mapping 21 MediaPipe hand landmarks to canvas operations (Draw, Erase, Clear, Hover) using Euclidean distance thresholds — runs fully client-side via WebAssembly with zero server overhead.
- Live: gesturelearn.vercel.app

Kadal Aayus — Sea Safety Mobile Application

Flutter · Firebase · Mapbox · FCM · Open-Meteo API

- Built a cross-platform sea safety app using Flutter and Firebase with real-time GPS telemetry streaming and Mapbox live fleet tracking.
- Integrated Open-Meteo API with an offline caching layer for 6-day weather matrices and localized Malayalam TTS alerts for low-connectivity coastal use.
- Programmed emergency SOS protocols with FCM push notifications for real-time hazard alerts to fleet members.

SKILLS

AI & Core: Generative AI, Agentic AI (LangGraph), Advanced RAG, Prompt Engineering, MCP

Languages: Python, Java

Frameworks: FastAPI, React, Node.js, Flutter

Databases: MySQL, ChromaDB, Weaviate

Tools & Infra: Docker, AWS Bedrock, Git, WebRTC, Firebase

EDUCATION

Master of Computer Applications (MCA)

2024 – 2026

TKM College of Engineering, Kollam

B.Sc. Computer Science

2020 – 2023

College of Applied Sciences, Vattamkulam

CERTIFICATIONS

IBM Generative AI & AI Agents Specialist — Coursera (IBM), 2026

AWS Generative AI and AI Agents Professional — Coursera (AWS), 2026

IBM Python for Data Science & Development — Coursera (IBM), 2026

Data Structures and Algorithms Design — NPTEL – IIT Kanpur, 2025

Python – Web Development Expert — NACTET, 2024